

Hierarchical Nitrogen-Doped Graphene/Carbon Nanotube Composite Cathode for Lithium–Oxygen Batteries

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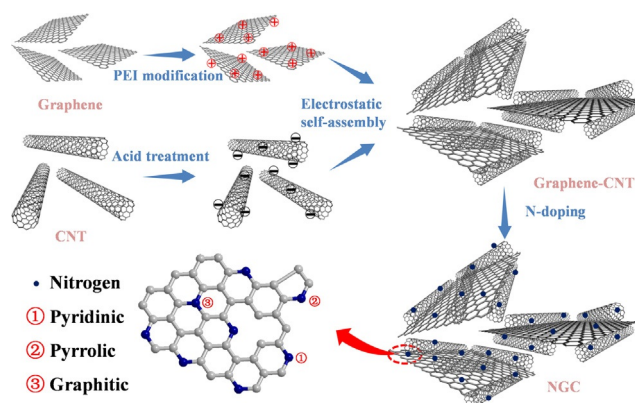
The lithium–oxygen (Li–O₂) battery is a very appealing candidate for advanced high energy applications owing to its exceptionally high specific energy. However, its poor energy efficiency, rate capability, and cyclability remain key barriers to its practical application. In this work, using a rationally designed cathode based on a bimodal mesoporous nitrogen-doped graphene/carbon nanotube (NGC) composite, we have developed a Li–O₂ battery demonstrating enhanced round-trip efficiency (ca. 85 %) and excellent cyclability over 400 cycles under a high current rate of 500 mA g^{−1}. The excellent cyclability and rate capability are attributed to improved stability of the aggressive LiO₂ intermediate on the nitrogen-doped carbon surface in addition to the favorable hierarchical architecture of NGC. These results demonstrate a valuable research direction to achieve highly stable and reversible Li–O₂ batteries through tuning the surface chemistry of the cathode in addition to finding a stable electrolyte solvent.

Rechargeable lithium–oxygen (Li–O₂) batteries have recently captured worldwide attention owing to their 5–10 times higher specific energy compared with Li-ion batteries.^[1] However, applications of Li–O₂ batteries still face challenges including inferior energy efficiency, poor rate capability, and short cycle life.^[2] These drawbacks are a result of the sluggish cathode reaction kinetics and the formation of indissoluble Li₂O₂. To release the potential of Li–O₂ batteries, it is thus imperative to develop air cathodes with reasonable structure and excellent catalytic activity to improve the reaction kinetics, mass transport, and product accommodation. Non-carbonaceous materials (nanoporous gold, perovskite, TiC, STO, etc.)^[3] could improve cycling performance of Li–O₂ batteries, but they lead to significant trade-offs in specific energy and power capability.^[4] On the contrary, carbon has several potential materials properties advantages that make it a good candidate as a cathode for Li–O₂ batteries (tailorable surface chemistry and structure, good conductivity, lightweight, large surface area and porosity, abundant structural variety, and low cost) if the surface

chemistry and catalytic activity could be improved and the structure could be carefully fabricated.^[5]

Herein, considering the surface chemistry and structure of cathode materials, we report the facile electrostatic self-assembly of graphene and carbon nanotubes (CNT) to obtain a 3D hierarchical bimodal mesoporous nitrogen-doped graphene/CNT composite (NGC) cathode, enabling a large number of catalytic active sites, fast reactant diffusion, sufficient electron conductivity, and accommodation of discharge products. The NGC cathode exhibited high round-trip efficiency, improved rate capability, and excellent cycling stability owing to a relatively large surface-to-volume ratio and reasonable pore structure distribution. To further explore the origin of the excellent electrochemical performance of NGC, the effect of surface chemistry of nitrogen-doped carbon on the cathode reaction pathway was also studied using density functional theory (DFT) calculations.

NGC was fabricated following the procedure shown in Scheme 1. Specifically, self-assembly between the poly(ethylene imine) (PEI)-modified graphene (zeta potential of +40 mV) and the oxidized CNTs (zeta potential of −38 mV) was achieved by electrostatic interaction. The powder obtained



Scheme 1. Fabrication process of NGC.

after evaporation of the solution was then heat-treated under argon atmosphere to obtain the graphene/CNT composite. Next, the as-prepared graphene/CNT composite was heated with dicyandiamide under argon atmosphere to dope nitrogen atoms into the 3D frameworks of both graphene and the CNTs to form NGC. TGA data confirms that PEI was completely removed after heat treatment (Figure S1 in the Supporting Information).

X-ray energy dispersive spectroscopy (EDS) elemental mapping (Figure 1 a) indicate that both C and N are homogeneously distributed in the as-synthesized NGC; most of the N atoms

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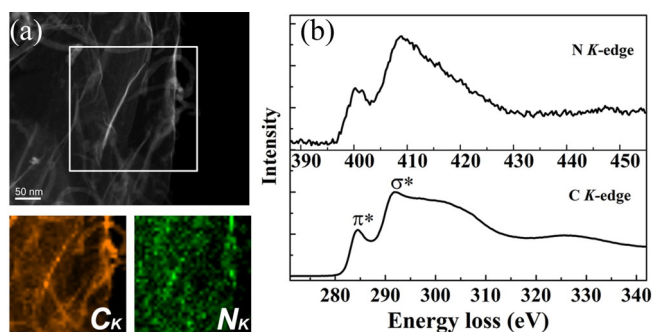


Figure 1. (a) STEM image and corresponding EDS elemental mapping of C and N. (b) N K-edge and C K-edge EELS spectra of NGC.

are bonded with sp^2 C atoms (pyridinic-N and pyrrolic-N) [X-ray photoelectron spectroscopy (XPS), C1s and N1s spectra, Figure S2c–d]. The presence of pyridinic and pyrrolic nitrogen in the nanocarbon materials were demonstrated to be efficient catalytic sites for oxygen electrochemistry in fuel cells^[7] and Li–O₂ batteries.^[8] The two similar bands starting at 280 and 395 eV in C K-edge and N K-edge electron energy loss spectroscopy (EELS, Figure 1b) are assigned to the transition of the $1s\text{-}\pi^*$ antibonding orbital, followed by another band attributed to the $1s\text{-}\sigma^*$ antibonding orbital.^[9] This type of EELS edge structure further corroborates the presence of sp^2 -hybridized bonds.^[10]

The NGC composite exhibits 3D hierarchical structure as observed by scanning electron microscopy (SEM, Figure S3a). CNTs are uniformly assembled on the surface of graphene sheet rather than agglomerated as shown in Figure S3c. The strong interaction of the interconnected CNTs and graphene sheets is supported by the upshift in the D band for NGC in the Raman spectra (1344 vs. 1335 cm^{-1} for graphene, Figure S2a).^[6] The XRD pattern (Figure S2b) of NGC demonstrates a more broadened diffraction peak than graphene, indicating smaller aggregation of graphene sheets and well-developed porosity in the NGC structure.^[6] The assembled CNT on graphene ideally results in a well-integrated 3D architecture and prevents sheet restacking; in this way, the availability of catalytic sites can be maximized. The porous structure of NGC is observed in the scanning transmission electron microscopy (STEM) image (Figure S3b) and N₂-sorption isotherm (Figure S4) that confirms the bimodal mesoporous structure of the NGC centered at 3.5 and 50 nm, which is the same as that of graphene/CNT. The measured surface area of the resulting material is $452\text{ m}^2\text{ g}^{-1}$. The noticeable pore size distribution and high surface area are highly desirable and essential for Li–O₂ battery operation because this type of architecture is appropriate both for the accommodation of a large amount of Li₂O₂ and for the efficient mass transport of the electrolyte and oxygen without severe blockage by Li₂O₂.

To evaluate the effect of nitrogen-doping and the 3D structure on battery performance, the discharge/charge curves of Li–O₂ batteries with NGC cathodes were compared with those of the Li–O₂ battery with undoped CNT and graphene cathodes. As seen in Figure S5, the NGC cathode exhibited the highest capacity (over 8000 mAh g^{-1}) and the lowest overpotential

compared with the other electrodes. The outstanding discharge capacity and the low overpotential of the NGC cathode are attributed to the favorable bimodal mesoporous structure and the increased O₂ adsorption and activation sites present after nitrogen doping.^[7,11] It should be noted that the cathodes are tested under obviously high current density (500 mA g^{-1}) compared to other carbon materials (200 mA g^{-1}),^[5c,d] demonstrating excellent rate capability of Li–O₂ batteries using NGC cathodes.

The cyclability of a Li–O₂ cell using a NGC cathode was tested under current density of 500 mA g^{-1} at a fixed specific capacity of 1000 mAh g^{-1} . As is shown in Figure 2, the cell

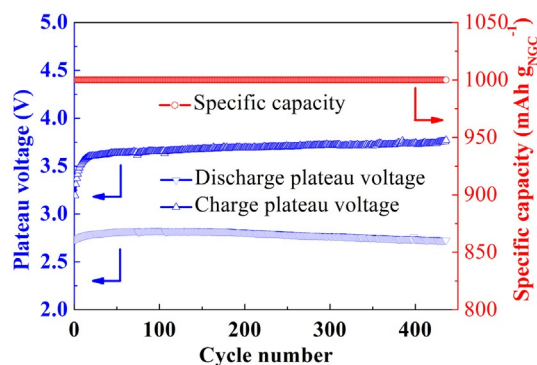


Figure 2. Cycling performance of the NGC cathode at a current density of 500 mA g^{-1} with a capacity of 1000 mAh g^{-1} .

based on NGC showed a constant capacity and low overpotential over 400 cycles and the initial round-trip efficiency reached up to 85%. To our knowledge, this value represents the highest cyclability and round-trip efficiency obtained on a carbon-based cathode (Figure S6e–g); it is also higher than the results reported for platinum-based catalysts.^[12] Even beyond the 400th cycle, the round-trip efficiency reached 73% with the charge plateau voltage remaining below 3.75 V. This is far below the decomposition potential of the electrolyte (4.5 V vs. Li⁺/Li, Figure S6a). Figure S6c and d present the cyclability of a NGC-based Li–O₂ battery under test runs at a capacity of 1000 mAh g^{-1} and current density of 1000 mA g^{-1} and 2000 mA g^{-1} . The battery maintained 100% capacity retention and coulombic efficiency after 200 and 78 cycles, respectively. The cyclability and rate capability of the NGC cathode far exceeded those of the non-carbonaceous electrodes reported previously, of which a comprehensive comparison is shown in Table S1.

The cathode process was investigated further with SEM and TEM analysis. After discharge, unlike the undoped electrode on which a dense, thick bulky layer typically forms (Figure S7b–d), uniform thin discharge products were observed on NGC (Figure 3a, Figures S7a and S8a). Formation of this uniform discharge product indicates that the oxygen reduction reaction (ORR) kinetics largely yield preferential growth of Li₂O₂ on nitrogen-doped carbon surfaces rather than Li₂O₂. The lower nucleation barrier for Li₂O₂ on NGC is likely dependent on the nature of the electrode surface (chemical and structural),

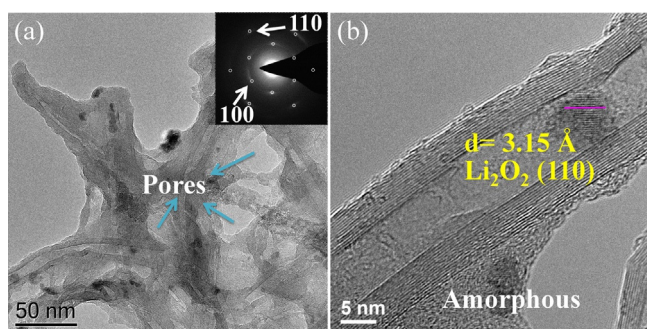


Figure 3. (a) TEM and (b) HRTEM images of the NGC cathode after the 10th discharge. Inset in (a): corresponding SAED of the NGC cathode after the 10th discharge. White circles are simulated Li₂O₂ [001] axis patterns, which are superimposed over the experimental diffraction pattern.

which is confirmed by DFT calculation (vide infra). Small nanoparticles and amorphous Li₂O₂ were identified by high-resolution TEM (HRTEM) in Figure 3b and S8b. The indexed selected area electron diffraction (SAED) pattern based on JCPDS #73-1640, shown in the inset of Figure 3a, indicates some crystalline Li₂O₂ nanoparticles existed on the cathode surface. The acquired Li and O K-edge EELS spectra of discharge product Li₂O₂ on NGC (Figure S8c–d) is different from that of commercial bulk Li₂O₂, which can be attributed to a large contribution of the surfaces and defects of electrochemically formed Li₂O₂ crystallites in NGC.^[13] The structural defects are important for improving electron transport in Li₂O₂,^[14] which is another reason for high rate capability of the Li–O₂ battery with a NGC cathode.

The ability to electrochemically recharge is necessary, but not sufficient proof for reversible Li₂O₂ formation/decomposition during discharging/charging.^[15] Only combining electrochemistry measurements with spectroscopic techniques can determine true reversibility. The FTIR spectrum of the NGC cathode (Figure S9a) demonstrates the formation and decomposition of Li₂O₂ after the 10th discharge and charge, indicating perfect reversibility of the Li–O₂ battery using a NGC cathode. However, the FTIR spectrum on an undoped graphene cathode shows that the discharge products contain Li₂CO₃ and organic carboxylate groups; the peak of Li₂O₂ may be overlapped by that of Li₂CO₃. After the 10th charge, residues of all discharge byproducts on undoped graphene cathode were observed. The byproducts are more difficult to be oxidized, and thus accumulate on the cathode during cycling, leading to observed capacity fading and premature cell death. The cathode was washed with D₂O and the solution was subjected to ¹H NMR analysis (Figure S10). Peak assignments reveal the presence of HCO₂Li and CH₃CO₂Li on the undoped graphene cathode, but not on NGC cathode after the 10th discharging/charging. The lack of any detectable HCO₂Li and CH₃CO₂Li evolution on NGC confirms that the amount of electrolyte decomposition occurring on the NGC cathode was smaller than on the undoped graphene cathode. XPS results also indicate that Li₂O₂ (54.7 eV) formed during discharge and was reversibly decomposed during charge on the NGC cathode (Figure S9b). These results verify that the NGC cathode can inhibit the generation of by-

products and favor the reversible formation/decomposition of Li₂O₂.

The improved cyclability observed for NGC cathode is in fact related to the surface chemistry, which is critical in determining the reaction mechanism at the cathode. It is generally accepted that O₂ is firstly reduced to O₂^{•−} through one electron reduction on the cathode, and then O₂^{•−} reacts with Li⁺ rapidly forming LiO₂. The succeeding predominant pathways are derived either from the further electrochemical reduction of LiO₂ or their chemical disproportionation through a solution-phase or a surface-migration route that forms Li₂O₂.^[17] Actually, LiO₂ is an aggressive intermediate that strongly influences the following cathode reaction processes. The decomposition of solvent molecules by LiO₂ through either nucleophilic substitution or proton abstraction reaction hampers the cyclability of Li–O₂ batteries.^[18] Thus, the stability of LiO₂ plays an important role in improving cyclability.^[19] To sufficiently understand the relationship between the surface chemistry of NGC and battery performance, the adsorption stability issues of LiO₂ on the surface of undoped and nitrogen-doped carbon were explored using DFT calculations.

We performed DFT calculations to follow the adsorptions of LiO₂ on pyrrole-doped graphene, pyridine-doped graphene, and undoped graphene. The most stable adsorbed structures and the adsorption energies of LiO₂ are shown in Figure S11. Interestingly, the stability of LiO₂ is enhanced after the nitrogen doping. The mechanism of the ORR at the cathode in Li–O₂ batteries is represented in Figure 4 by using the following equations:

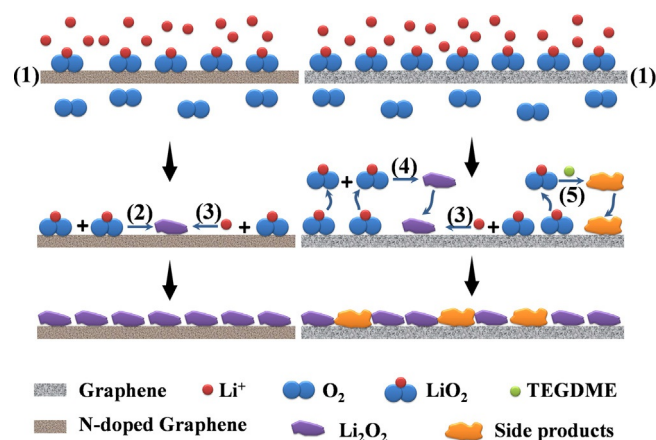
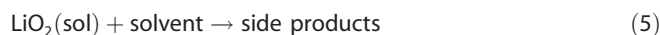
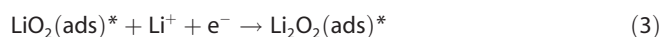
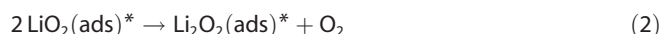


Figure 4. Illustration of the ORR mechanism on different cathode surfaces.



where (ads) refers to the molecules adsorbed onto the surface and (sol) refers to the solvated molecules. The high adsorption

energy of LiO_2 on nitrogen-doped graphene implies that the succeeding electrochemical reduction and disproportionation of LiO_2 may proceed through surface diffusion and migration [Figure 4, Eq. (2) and (3)]. A weak adsorption of LiO_2 on undoped carbon may lead to dissolution of LiO_2 and thus result in a solution reaction forming Li_2O_2 that can re-deposit on the cathode surface [Figure 4, Eq. (4)]. Overall, the nitrogen doping stabilizes the LiO_2 relative to the undoped cathode. Stabilization of LiO_2 on nitrogen-doped graphene inhibits the occurrence of nucleophilic substitution and proton abstraction side reactions between dissolved LiO_2 and the solvent molecules [Figure 4, Eq. (5)], which is crucial to inhibit the formation of side products. Experimental results are consistent with the mechanism shown in Figure 4.

As shown in Figure S7a and e, the uniformly thin products on NGC were formed after discharge and the electrode structure was regained after charge. In contrast, the bulk particles, which were considered as the typical products produced through solution reaction process,^[20] were observed on the undoped graphene/CNT cathode after discharge and the bulk film was not removed after charge owing to the presence of side products (Figure S7d and f). Gas chromatography (GC) analyses (Figure S12) showed only O_2 was released from the cell with NGC, whereas both O_2 and CO_2 were found from the cell with the undoped cathode during charging. These results further corroborate the inhibition of side reactions by using nitrogen-doped cathodes. Stabilization of LiO_2 on the cathode surface is superior than in the electrolyte solvent reported recently^[19] because the direct reaction between LiO_2 and the Li anode is inhibited. Thus, incorporating N species on the carbon surface to stabilize aggressive intermediates and inhibit the reaction between the intermediates and electrolyte solvent could be a valuable strategy to improve the intrinsic cyclability of Li– O_2 batteries.

In summary, a Li– O_2 battery with high round-trip efficiency, cyclability, and rate capability was demonstrated using a bimodal mesoporous nitrogen-doped graphene–carbon nanotube composite cathode. The excellent performance was attributed to the bimodal mesoporous structure facilitating the mass transport kinetics in the electrochemical reactions and the improved stability of the cathode reaction intermediate LiO_2 on the carbon surface after introduction of nitrogen atoms inhibiting the unwanted side reaction. This work clearly demonstrates that the performance of Li– O_2 batteries could be dramatically improved by rationally designing the electrode structure and tuning the surface chemistry of the electrode material. Notably, investigation of the relationship between the stability of LiO_2 and the surface chemistry of the cathode material demonstrated here point a new valuable direction for improvements in battery stability.

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